Putting It All Together

Causality, DiD, Panel Data, RDD — a midterm recap

Lecture 5

Today's roadmap

This lecture is a *review* — no new core methods, but a few new **linking ideas** that tie the toolkit together.

1. The one question all four methods answer
2. DAGs: a unifying language for selection bias
3. DiD with multiple time periods (bridging DiD and panel data)
4. Side-by-side: counterfactuals, assumptions, estimands
5. How the methods nest inside each other
6. Common pitfalls (and how to recognise them on an exam)
7. A worked example
8. A peek ahead: instrumental variables

The one question

Every causal method answers the same question

The fundamental problem
We want $\mathbb{E}[Y_i(1) - Y_i(0)]$ but only observe one potential outcome per unit.

What we *can* compute — the naive comparison — decomposes as:

$$\underbrace{\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]}_{\text{observed}} = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT (causal)}} + \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}$$

Every method we have learned is a strategy for killing the second term.

Four strategies, one goal

Method	**How it kills selection bias**	**Counterfactual**
RCT	Random assignment $\Rightarrow$ groups balanced	The control group
DiD	Selection bias cancels in changes	Control's *trend*
Panel (FE/FD)	Differences out unit-specific traits	Unit's own past
RDD	Local randomization near cutoff	The other side of c_0

The toolkit isn't four unrelated tricks. It's four ways of constructing a credible **counterfactual** when the treated and control groups aren't directly comparable.

A unifying language: DAGs

Why DAGs?

We have been speaking informally about "omitted variables" and "confounders." Directed acyclic graphs (DAGs) make this precise.

A DAG is a picture with:

- **Nodes**: variables
- **Arrows**: direct causal effects
- No cycles (no variable causes itself)

Useful because they let you *read off* which variables you must control for, which you must *not* control for, and what kind of bias you have.

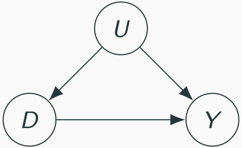
(Cunningham, Mixtape, Ch. 3 — recommended reading.)

The three building blocks

Chain

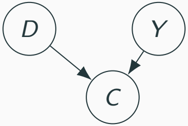
M is a *mediator*.
Don't control for it if you want the total effect of D.

Fork (confounder)



U is a *confounder*.
Must control for it.

Collider

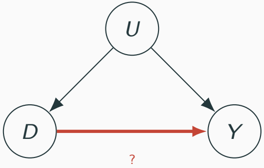


C is a *collider*.
Don't control for it — doing so creates bias.

The deep idea: a "backdoor path" from D to Y (any non-causal route) biases the OLS estimate.
Close every open backdoor.

Each method, drawn as a DAG

The selection-bias problem



U (unobserved) opens a backdoor. OLS of Y on D is biased.

How each method blocks the backdoor

- **RCT**: cut the $U \to D$ arrow by randomising D.
- **Panel FE**: if $U = a_i$ is time-invariant, difference it out.
- **DiD**: if U shifts levels but not trends, take differences over time *and* across groups.
- **RDD**: near c_0, U is (assumed) continuous, so its effect on Y doesn't jump — only D does.

DiD with multiple time periods

Where we are in the toolkit

We have seen two "DiD-ish" worlds:

- **Canonical DiD** (Lecture 2):
  2 groups, 2 periods, one δ coefficient.
  Examples: Card–Krueger NJ vs. PA minimum wage.
- **Two-way fixed effects** (Lecture 3):
  N units, T periods, one β on a treatment dummy D_{it}.
  Examples: state minimum-wage panels (Clemens & Strain).

Today's question: what lives *between* those two? Specifically:

- Many treated groups
- Many time periods
- Different groups become treated at *different* times

This is called **staggered treatment adoption**, and it turns out our workhorse TWFE estimator can break in subtle ways here.

Staggered adoption is the default in applied economics

Real-world examples where units are treated *at different times*:

- State-by-state minimum-wage increases (different states raise at different years).
- Country-by-country EU accession (Greece 1981, Spain/Portugal 1986, . . . , Croatia 2013).
- Firm-by-firm technology adoption (broadband, ERP systems, ChatGPT).
- Region-by-region rollout of a healthcare programme.
- Worker-by-worker entry into a training programme.

In all these cases:

- There is no single "treatment year."
- Each unit has its *own* pre/post boundary.
- Once treated, a unit usually stays treated ("staggered adoption" assumption).

The standard advice for decades: just run TWFE. Recent literature (2018–2021) says: *not so fast.*

Quick review: the canonical 2×2 case

Setup. Two periods $t-1$ (pre) and t (post), two groups.

Potential outcomes notation:

- $Y_{is}(0)$ = outcome for unit i in period s *if not treated*.
- $Y_{is}(1)$ = outcome for unit i in period s *if treated*.

Observed outcomes (with no anticipation):

$$Y_{i,t-1} = Y_{i,t-1}(0), \qquad Y_{it} = D_i\, Y_{it}(1) + (1 - D_i)\, Y_{it}(0)$$

Parameter of interest — the ATT:

$$\text{ATT} = \mathbb{E}\big[Y_t(1) - Y_t(0) \,\big|\, D = 1\big]$$

Parallel trends assumption:

$$\mathbb{E}\big[Y_t(0) - Y_{t-1}(0) \,\big|\, D = 1\big] = \mathbb{E}\big[Y_t(0) - Y_{t-1}(0) \,\big|\, D = 0\big]$$

Generalising: the TWFE regression

Now let there be $\mathcal{T}$ total time periods. The standard tool is the **two-way fixed effects** regression:

$$\boxed{Y_{it} = \theta_t + \eta_i + \alpha D_{it} + v_{it}}$$

The pieces:

- θ_t = time fixed effect (one dummy per period; absorbs common shocks).
- η_i = unit fixed effect (one dummy per unit; absorbs time-invariant traits).
- D_{it} = treatment dummy: 1 if unit i is treated by time t, else 0.
- v_{it} = idiosyncratic shock, assumed mean-independent of the rest.
- α = the parameter we hope is "the treatment effect."

The standard interpretation: α is the average effect of being treated.

The problem we'll spend the rest of the section on: that interpretation breaks down when treatment timing varies *and* effects are heterogeneous.

When does TWFE actually deliver the ATT?

TWFE gives a clean, interpretable ATT in exactly two scenarios:

Case 1: Treatment effects are truly homogeneous.
If $Y_{it}(1) - Y_{it}(0) = \alpha$ for *every* unit i in *every* period t, TWFE recovers α. But homogeneity is a strong assumption rarely defensible in applied work.

Case 2: There are only two time periods.
The canonical case from Lecture 2. With one treated group, one untreated group, and one pre/one post period, $\alpha =$ ATT exactly under parallel trends and no anticipation.
Even with heterogeneous effects, α recovers their average.

The key insight: the robustness of TWFE in the 2-period case is a *coincidence*, not a general property. It does *not* extend to multi-period staggered settings.

The leading case in applied work: many periods, staggered adoption, heterogeneous effects across cohorts and over time. *This is where TWFE breaks.*

Why “heterogeneous effects” is the leading case

In applied work, treatment effects almost always vary along at least one of:

- **Across units.** The minimum wage probably affects fast-food employment in Mississippi differently than in California.
- **Across cohorts.** Early adopters of a policy may differ from late adopters — they self-selected into early adoption.
- **Over elapsed time (dynamics).** A training programme's effect may build slowly, peak, then fade.
- **Across calendar time.** A subsidy enacted in a boom may have different effects than one enacted in a recession.

Implication: if you write $Y_{it}(1) - Y_{it}(0) = \alpha_{it}$ (a unit- and time-specific effect), then summarising everything with a single α is a modelling choice, not a fact.

The question is: when you run TWFE, *which* weighted average of the α_{it}'s does $\hat{\alpha}$ actually estimate? The answer turns out to be: a weighted average where some weights can be

The core problem: TWFE's implicit comparisons

TWFE estimates α by comparing units whose treatment status *changes* to units whose status *does not change* (within a time period).

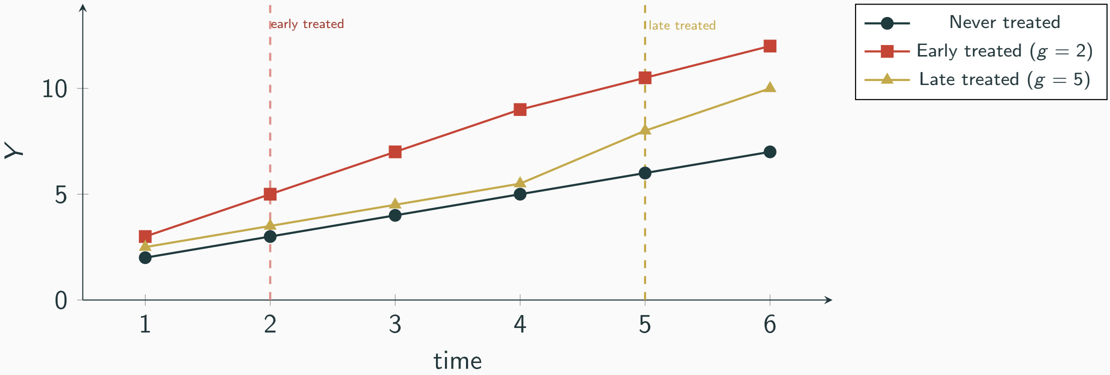
With staggered timing, in any given period t, the "not changing" group contains:

Comparison group	**Verdict**
Newly treated vs. **never treated**	**Good**
Newly treated vs. **not-yet treated**	**Good**
Newly treated vs. **already treated**	**Bad!**

The first two are in the spirit of DiD — they compare changes in treatment to changes in *untreated* potential outcomes.

The third compares newly-treated units to units that *are already treated*. Their trajectories include treatment-effect dynamics, not $Y(0)$.

The forbidden comparison, in pictures



When TWFE evaluates the late group's treatment effect at $t = 5$, it partly uses the early group as a control. But the early group's outcomes at $t = 5$ already include *three periods* of treatment dynamics — that's not $Y(0)$.

The contamination shows up inside $\hat{\alpha}$, in a way you can't see from the regression output

How bad can it get? The negative-weights problem

Goodman-Bacon (2021) decomposes $\hat{\alpha}_{\mathsf{TWFE}}$ into a weighted sum of all possible 2×2 DiD comparisons between cohorts.

Two key results from this decomposition:

1. Treatment dynamics get subtracted.
When an already-treated cohort is used as the control, its treatment-effect trajectory enters the comparison *with the wrong sign*.

2. Weights can be negative.
The implicit weights on individual ATTs are determined by group sizes and treatment timing. They can be negative for some cells.

The worst-case consequence:

> *It is possible to construct examples where the treatment effect is positive for every unit in every period, yet $\hat{\alpha}_{TWFE} < 0$.*

Even when negative weights are ruled out, $\hat{\alpha}$ recovers a weighted average of ATTs with weights

The Bacon decomposition: TWFE as a weighted average of 2×2s

With two treated cohorts (A = early, B = late) and never-treated (U), TWFE implicitly averages *four* 2×2 DiDs:

2×2 comparison	Status
A treated vs. U never	Good — U is a clean control.
B treated vs. U never	Good — U is a clean control.
B treated vs. A not-yet (pre-A window)	Good — A hasn't been treated yet.
B treated vs. A *already* treated (post-A window)	Bad — A's outcomes contain dynamics.

TWFE's $\hat{\alpha}$ = sum of these four DiDs, weighted by group sizes and treatment-window variance.

The last row is the troublemaker. If A's treatment effect is still evolving when B gets treated, the contamination shows up in $\hat{\alpha}$.

The remedy: don't pool, disaggregate

Callaway–Sant'Anna (2021) propose a clean fix: stop trying to summarise everything in one regression. Instead, estimate a *separate* ATT for each (cohort, period) cell.

New notation:

- G_i = the time period when unit i *first* becomes treated (the "cohort" or "group").
- $C_i = 1$ if unit i is in the never-treated group.
- $Y_{it}(g)$ = potential outcome in period t for a unit first treated in period g.
- $Y_{it}(0)$ = potential outcome if *never* treated.

The group-time average treatment effect:

$$\boxed{\mathrm{ATT}(g,t) = \mathbb{E}\big[\, Y_t(g) - Y_t(0) \,\big|\, G = g \,\big]}$$

In words: the average causal effect of being treated, for the cohort that was *first* treated in period g, evaluated $t - g$ periods after they got treated.

Sanity check: in the canonical case (treated in period 2, only 2 periods), this collapses to

ATT(g, t): concrete examples

Suppose you have data for 3 periods. Two cohorts get treated: one in period 2, one in period 3.

Parameter	What it measures
ATT$(g{=}2,\ t{=}2)$	Effect for the cohort treated in period 2, on impact (in period 2).
ATT$(g{=}2,\ t{=}3)$	Effect for the cohort treated in period 2, one period after treatment.
ATT$(g{=}3,\ t{=}3)$	Effect for the cohort treated in period 3, on impact.

What we don't compute: ATT$(g{=}3, t{=}2)$ — cohort 3 isn't yet treated at $t = 2$. Only define ATT(g, t) for $t \geq g$.

The output of a CS estimation is a matrix of $\widehat{\text{ATT}}(g, t)$'s, not a single number. Each cell is a transparent 2×2 DiD with a clean comparison group.

We can then choose how to aggregate them — with full control over what we're averaging.

Required assumption 1: staggered treatment adoption

Staggered adoption: for all $t = 1, \ldots, \mathcal{T} - 1$,

$$D_{it} = 1 \implies D_{i,t+1} = 1.$$

In plain English: **once treated, always treated.** Units don't "un-treat."

When this holds naturally:

- Policies that roll out and stay in place (e.g. legalisation, deregulation).
- "Scarring" treatments: ever-participating in job training defines treatment status.
- Technology adoption (once a firm has broadband, it has broadband).

When this fails:

- Treatments that turn off (e.g. short-run subsidies, seasonal programmes).
- On/off interventions like trade tariffs that get repealed.

For non-staggered settings, things get harder — you typically need additional restrictions on heterogeneity. The CS framework focuses on staggered as the leading case.

Required assumption 2a: parallel trends (never-treated)

For every cohort $g \geq 2$ and every period $t \geq g$:

$$\boxed{\mathbb{E}\big[Y_t(0) - Y_{t-1}(0) \,\big|\, G = g\big] = \mathbb{E}\big[Y_t(0) - Y_{t-1}(0) \,\big|\, C = 1\big]}$$

In plain English: in the absence of treatment, the cohort first treated in period g would have followed the same path as the never-treated group.

What's different from the canonical case:

- This needs to hold for *every* cohort separately.
- It needs to hold in *every* post-treatment period — not just on impact.

When you can use it: you have a (sufficiently large) never-treated group that is plausibly similar to the eventually-treated cohorts.

When you can't: if everyone eventually gets treated (e.g. all 50 states adopted), there's no never-treated group. Use the next variant.

Required assumption 2b: parallel trends (not-yet-treated)

The alternative is to compare cohort g to all units that haven't *yet* been treated by period s. For $g \geq 2$, $t \geq g$, $s \geq t$:

$$\mathbb{E}\left[Y_t(0) - Y_{t-1}(0) \,\middle|\, G = g\right] = \mathbb{E}\left[Y_t(0) - Y_{t-1}(0) \,\middle|\, D_s = 0,\; G \neq g\right]$$

Intuition: use the late cohorts as controls for the early cohorts, *before* the late cohorts become treated.

Pros:

- Uses more data (more comparison units in early periods).
- Works even when there is no never-treated group.

Cons:

- Slightly stronger: parallel trends now restricts comparisons *across cohorts*, not just to a never-treated benchmark.
- Marcus & Sant'Anna (2021): "there is no free lunch."

Identification: how to estimate ATT(g, t)

Under parallel trends with a **never-treated** comparison group:

$$\boxed{\text{ATT}(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]}$$

This is just a 2×2 DiD, using:

- **Pre-period:** $g - 1$, the period just before cohort g's treatment.
- **Post-period:** t, the period we're evaluating.
- **Treated:** cohort g.
- **Control:** the never-treated group.

With **not-yet-treated** comparison instead:

$$\text{ATT}(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0,\ G \neq g]$$

The key takeaway: every $\widehat{\text{ATT}}(g,t)$ is a transparent 2×2 DiD with a clean control group — no contamination from already-treated units.

Conditional parallel trends: when covariates matter

Sometimes parallel trends is implausible *unconditionally* but plausible *after conditioning* on observables.

Example. You're studying the effect of job training on earnings. Earnings trends depend on baseline education, occupation, experience. If training participants have different distributions of these variables, you need to condition on them.

Conditional parallel trends (never-treated version):

$$\mathbb{E}\left[Y_t(0) - Y_{t-1}(0) \,\middle|\, X, G = g\right] = \mathbb{E}\left[Y_t(0) - Y_{t-1}(0) \,\middle|\, X, C = 1\right]$$

Estimation in practice — three equivalent (under correct specification) approaches:

1. **Outcome regression:** model $\mathbb{E}[\Delta Y \mid X]$ for the control group.
2. **Inverse probability weighting:** reweight controls to match the cohort's X distribution.
3. **Doubly-robust:** combine the two; consistent if *either* model is right.

The R `did` package implements all three and defaults to doubly-robust.

Aggregating $\text{ATT}(g,t)$'s: the menu

With G cohorts and many post-treatment periods, you may have dozens of $\widehat{\text{ATT}}(g,t)$'s. How do you summarise?

The CS framework offers **three principled aggregations**, each answering a different question:

Aggregation	**Question it answers**	**Notation**
By cohort	"How does the effect vary by who got treated when?"	$\theta_S(g)$
By elapsed time	"How do effects evolve with time-since-treatment?"	$\theta_D(e)$
Overall	"What's the single-number summary?"	θ_S^O

We'll go through each in turn. The good news: all three are simple weighted sums of the underlying $\text{ATT}(g,t)$'s, with weights you choose deliberately.

Aggregation 1: effect by cohort

The average effect for cohort g, taken over all post-treatment periods:

$$\boxed{\theta_S(g) = \frac{1}{\mathcal{T} - g + 1} \sum_{t=g}^{\mathcal{T}} \mathrm{ATT}(g, t)}$$

In words: for the cohort first treated in g, the average treatment effect across all the post-periods we observe for them.

When this is interesting:

- You suspect early adopters and late adopters differ (selection into early adoption).
- You want to test whether the policy's effectiveness depended on when it was implemented (e.g. rolled out in a boom vs. a recession).

What it looks like in practice: plot $\hat{\theta}_S(g)$ against g. A flat line means homogeneous effects across cohorts; a slope tells you something about timing-related heterogeneity.

Note: for each cohort, we average over a *different number* of post-periods (later cohorts have

Aggregation 2: the overall ATT

If you must report a single number, the CS-recommended summary is:

$$\boxed{\theta_S^O := \sum_{g=2}^{\mathcal{T}} \theta_S(g)\, P(G=g)}$$

In words: average the cohort-level effects $\theta_S(g)$, weighting each by the population share of that cohort.

Why this is the right "analog of the ATT":

- It weights each cohort by how big it is (not by some odd function of treatment timing, as TWFE does).
- Every weight is non-negative.
- The interpretation is clean: "the average effect of the policy across all units that were ever treated."

Reporting recommendation. If the journal/referee insists on one headline number, report θ_S^O.

Aggregation 3: the event study by elapsed time

The most popular aggregation in applied work — it shows treatment-effect dynamics.

Define elapsed time $e = t - g$ (periods since treatment). Then:

$$\boxed{\theta_D(e) := \sum_{g=2}^{\mathcal{T}} \mathbf{1}\{g + e \leq \mathcal{T}\}\, \mathrm{ATT}(g, g+e)\, P(G = g \mid g + e \leq \mathcal{T})}$$

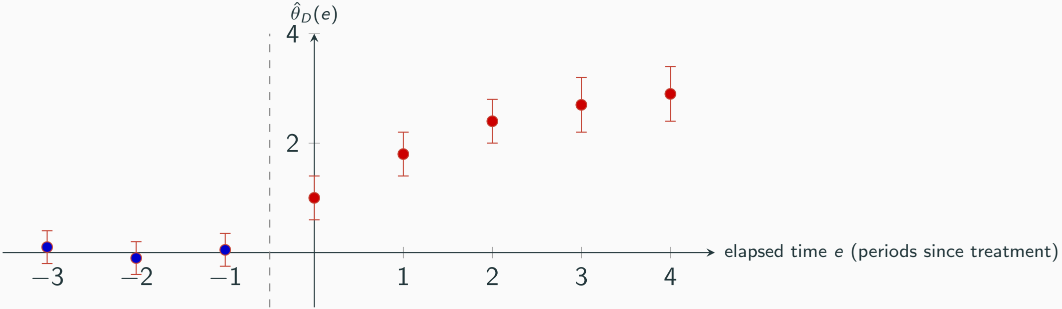
In words: the average treatment effect e periods after treatment, across all cohorts for whom we observe that elapsed period.

What you do with it: plot $\hat{\theta}_D(e)$ against e.

- $e = 0$: the on-impact effect.
- $e > 0$: dynamic effects — does the effect build up? Fade out?
- $e < 0$: *pre-treatment placebo effects* — should be near zero if parallel trends holds.

This is the clean, modern replacement for the lead/lag dummies in a TWFE event-study.

What a clean event-study plot looks like



What to look for:

- Pre-treatment estimates clustered near zero with confidence intervals containing zero $\Rightarrow$ parallel trends looks credible.
- Post-treatment: shape tells you the dynamics — here, effects build up over 3 periods then plateau.

Workflow comparison: TWFE vs. Callaway–Sant'Anna

	Classical TWFE	**Callaway–Sant'Anna**
What you fit	One regression: α on D_{it}, with unit + time FE	A separate $2{\times}2$ DiD for each (g, t) cell
Comparison group	All untreated-at-t units (including already-treated)	Never-treated OR not-yet-treated
Output	One number, $\hat{\alpha}$	A matrix $\widehat{\text{ATT}}(g, t)$, plus chosen aggregations
Robust to heterogeneous effects?	No (in general)	Yes
Negative weights possible?	Yes	No
Event studies	Lead/lag dummies, susceptible to contamination	$\theta_D(e)$ aggregation, clean by construction
Covariates	Add as regressors	Conditional parallel trends + OR/IPW/DR
Software	`plm`, `fixest::feols`, Stata `reghdfe`	R: `did` package; Stata: `csdid`

Rule of thumb: TWFE for simultaneous treatment, CS for staggered treatment.

When TWFE is still fine (don't over-correct)

The modern DiD critique applies in specific circumstances. TWFE remains a sensible choice when:

1. **Treatment is simultaneous.**
   All treated units adopt at the same time. There is no "already-treated" contamination because there is no early/late distinction.
2. **Two periods, two groups.**
   The canonical Card–Krueger case. TWFE and ATT coincide.
3. **You have strong reasons to believe effects are homogeneous.**
   Rare in applied work, but possible (e.g. tightly-controlled lab settings).
4. **As a robustness check** alongside CS estimates — agreement between TWFE and CS is reassuring.

Don't panic-replace every TWFE in your dissertation. Replace TWFE where:

- Treatment timing varies, AND

Related families of estimators

Callaway–Sant'Anna (2021) is not the only modern DiD estimator. Three contemporaneous papers diagnose the same problem with different fixes:

- **Goodman-Bacon (2021).**
  Doesn't propose a new estimator; instead, *decomposes* TWFE into 2×2s so you can see what your $\hat{\alpha}$ is actually averaging. Diagnostic tool.
- **de Chaisemartin & D'Haultfœuille (2020).**
  Propose an estimator (DID_M) using only "switchers" — units that change treatment status between consecutive periods. Robust to negative weights.
- **Sun & Abraham (2021).**
  Specifically fixes the *event-study* version of TWFE. Their interaction-weighted estimator gives clean elapsed-time effects.
- **Borusyak, Jaravel & Spiess (2024).**
  Imputation-based approach: impute $Y(0)$ for treated observations using untreated ones.

Practical checklist for applied work

When confronted with a staggered DiD problem, here is the recommended workflow:

1. **Look at your treatment timing.** How many cohorts? Is there a never-treated group?
2. **Run the Bacon decomposition.** See how much weight TWFE puts on the bad "already-treated" comparisons. (Stata: `bacondecomp`.)
3. **Estimate CS group-time ATTs.** Use never-treated controls if available; otherwise not-yet-treated.
4. **Plot the event study $\hat{\theta}_D(e)$.** Check pre-treatment estimates for parallel trends.
5. **Report the overall ATT $\hat{\theta}_S^O$** as your headline.
6. **Show robustness:** cohort-by-cohort $\hat{\theta}_S(g)$, alternative comparison groups, alternative estimators (dCDH, S&A, BJS).
7. **Compare to TWFE** and explain any large divergence.

This is now standard practice at top journals (AER, QJE, JPE, ReStud, JoE).

TWFE is a tool, not a default.

Three things to remember:

1. Canonical 2×2 DiD is the foundation. Always start there conceptually.
2. TWFE generalises 2×2 DiD cleanly only when treatment is simultaneous or effects are homogeneous.
3. For staggered adoption with likely heterogeneous effects, switch to group-time ATTs — they are transparent, robust, and just as easy to estimate with modern software.

And a unifying thought: the entire modern DiD literature is one extended argument about the same question we've been asking all semester — *what is the right counterfactual?* The bad comparisons aren't bad because the math is hard; they're bad because "already-treated units" do not give you $Y(0)$.

Side-by-side comparison

The cheat sheet

	DiD	Panel FE / TWFE	Sharp RDD	Fuzzy RDD
Data needed	2+ groups, 2+ periods	N units $\times$ T periods	Cross-section + running var.	Same as sharp
Key assumption	Parallel trends	Strict exogeneity (given a_i)	Continuity of $\mathbb{E}[Y(d) \mid X]$ at c_0	Continuity + first-stage jump
What it kills	Time-invariant level differences	Time-invariant unobservables	Anything smooth at c_0	Same
Estimand	ATT (treated, post)	Within-unit effect	LATE at c_0	LATE for compliers at c_0
Headline regression	$Y = \beta_0 + \beta_1 \text{Post} + \beta_2 \text{Treat} + \delta(\text{Post} \cdot \text{Treat})$	$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + e_{it}$	$Y = \alpha + \tau D + f(X - c_0) + e$	2SLS using $\mathbf{1}[X \geq c_0]$ as IV
Main threat	Differential trends	Feedback (no strict exog.)	Manipulation / other policies at c_0	Same + weak first stage

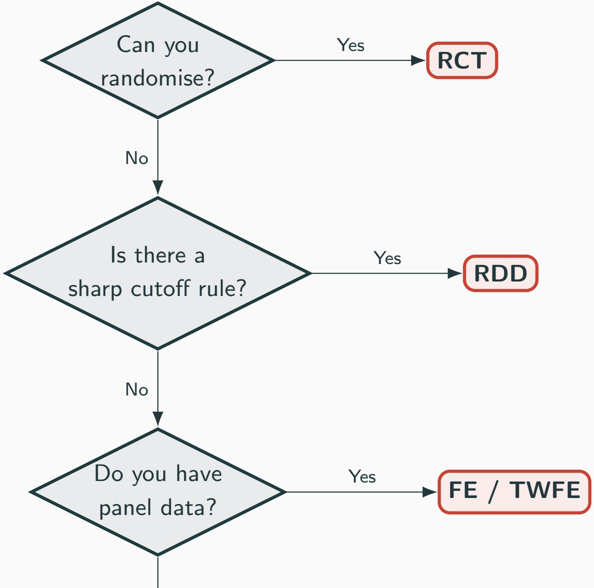
The nesting structure

The methods aren't separate islands. They nest:

$$\boxed{\begin{array}{c}\textbf{TWFE}\\ \text{(general panel)}\end{array}} \xrightarrow{N=2,\ T=2} \boxed{\begin{array}{c}\textbf{DiD}\\ \text{(2 groups, 2 periods)}\end{array}} \xrightarrow{\text{no time dim.}} \boxed{\begin{array}{c}\textbf{Pooled cross-section}\\ \text{(1 group, 1 period)}\end{array}}$$

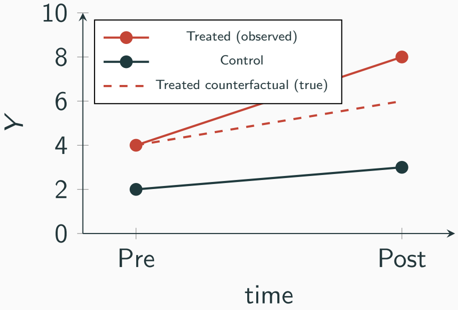
And RDD? It uses a single cross-section but exploits a different kind of variation: *local* randomization around a cutoff. So it sits orthogonally to the time-based methods — handy when you have no panel and no good control group, but you do have a rule.

Which method, when?



Common pitfalls

Pitfall 1: DiD when trends aren't parallel



DiD assumes the counterfactual gap mirrors the control trend.

If the treated group was already on a steeper path, DiD attributes the pre-existing divergence to treatment.

Defense: pre-trend tests, event studies, placebo periods.

Pitfall 2: Pooled OLS with panel data

Running OLS on stacked panel data ignores the unit-specific effect a_i:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \underbrace{(a_i + e_{it})}_{\text{composite error}}$$

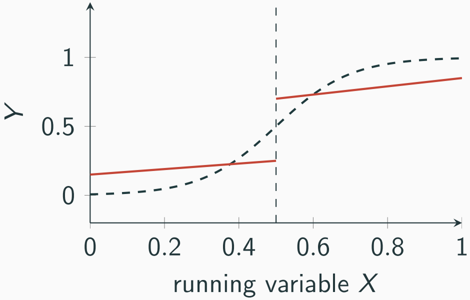
If X_{it} correlates with a_i (it almost always does in applied work) — bias.

The fix is to use within-unit variation:

- **FD**: $\Delta Y_{it} = \beta_1 \Delta X_{it} + \Delta e_{it}$ (good for sharp shocks)
- **FE**: demean by unit (better for gradual effects, more efficient)

Both kill a_i. The choice between them depends on *how* you think the effect plays out over time.

Pitfall 3: RDD — curvature mistaken for a jump



A genuinely smooth (but curved) $\mathbb{E}[Y \mid X]$ can look like a jump if you force linear fits on each side.

This is a kink, not a discontinuity.

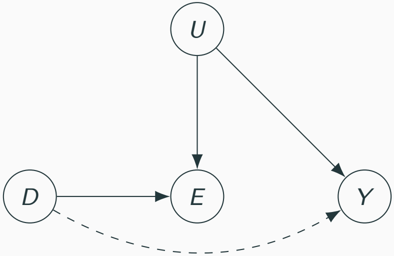
Defense:

- Plot the raw data first
- Try flexible polynomials *and* local linear fits
- Check robustness across bandwidths

Pitfall 4: Controlling for the wrong thing

A DAG mistake that bites in applied work:

Bad control = a collider. Suppose treatment D affects employment E, and ability U affects both wages Y and employment E:



If you condition on E, you open the path $D \to E \leftarrow U \to Y$ — selection bias appears *because* of your control.

Lesson: “more controls” isn't always better. Pretreatment variables only.

A worked example

Setup: a state-level training subsidy

Scenario. In 2018, three U.S. states introduced a wage subsidy for firms that hire workers passing a federal skills exam with score ≥ 60. You have firm-level data on hiring, 2015–2022, for all 50 states.

Question: does the subsidy raise hiring?

- Which method(s) apply?
- What's the identifying assumption in each case?
- What estimand do you recover?

Think for 30 seconds, then we'll go through it.

Three valid designs — not one

Design A: DiD across states.

Treated: the 3 adopting states; control: the other 47. Compare Δhiring 2015–17 vs. 2018–22.

Assumes: parallel trends in hiring absent the policy.

Estimand: ATT in adopting states.

Design B: Two-way fixed effects.

$\text{Hiring}_{it} = \alpha_i + \lambda_t + \beta\, \text{Subsidy}_{it} + e_{it}$

Assumes: strict exogeneity given α_i; identical effect across the 3 states.

Estimand: the (weighted) average within-state effect of the subsidy.

Caveat: the 3 states adopted simultaneously in 2018 here, so TWFE is fine. With staggered timing, use Callaway–Sant'Anna instead.

Design C: Fuzzy RDD on the score cutoff.

Within a treated state, compare firms whose marginal hire scored just below vs. just above 60.

Assumes: continuity of potential outcomes at score $= 60$; no manipulation of scores.

Estimand: LATE for compliers near the cutoff.

Why three answers can both differ and agree

The three estimands answer *different* questions:

- DiD/TWFE $\rightarrow$ average effect across all subsidised firms in adopting states
- RDD $\rightarrow$ effect for the *marginal* firm at score $= 60$

If treatment effects are heterogeneous, the methods will give different numbers — and **that's not a contradiction**, it's information.

The credibility revolution playbook:
Run multiple designs, ask whether they agree, and if they don't, ask *why* they don't.

Looking ahead

One thread we'll pick up later

Instrumental variables.
Fuzzy RDD is secretly a 2SLS estimator: the cutoff dummy $\mathbf{1}[X \geq c_0]$ is the instrument for treatment. IV is the general case of this idea.

Use when: you have something that shifts D but doesn't directly affect Y.

Examples we'll see:

- Distance to college as an instrument for years of schooling
- Lottery draws as instruments for program participation
- Quarter-of-birth as an instrument for schooling (Angrist & Krueger)

IV will close the loop on the toolkit: RCT, DiD, panel, RDD, and now IV — each is a different strategy for finding exogenous variation in D.

Midterm checklist

What you should be able to do

1. Write the selection-bias decomposition and identify each term.
2. Given a research question, choose an appropriate design and justify the choice.
3. State the key identifying assumption of DiD, FE, sharp RDD, and fuzzy RDD — in words *and* in math.
4. Read a regression specification and explain what each coefficient means (especially the DiD δ and the FE within transformation).
5. Diagnose what's wrong with a flawed design (bad control, broken parallel trends, manipulation at the cutoff, ...).
6. Distinguish ATE, ATT, LATE, and ATT(g, t) — and say which you've estimated.
7. Connect the 2$\times$2 DiD table to the TWFE regression, and explain why TWFE can fail with staggered adoption.

Causal inference = building credible counterfactuals.

Every method we've covered is one way of building that counterfactual.

The art is choosing the one whose assumptions you can defend in *your* setting.

Good luck on the midterm.

References

- Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale UP. Chs. 3, 6, 8, 9.
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